

Customer Churn Analysis & Insights Report

Telco Customer Churn Dataset | IBM Sample Data

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1. Introduction

In today's competitive telecom market, keeping customers is one of the most important things a company has to do. Customers leave for all kinds of reasons, maybe the price is too high, maybe they found a better plan somewhere else, or maybe they just weren't happy with the service. This is called customer churn, and it is a big problem because getting a new customer costs way more than keeping an existing one.

The main goal of this project was to analyze a telecom company's customer data and figure out, who is leaving and more importantly, why are they leaving. I wanted to find patterns in the data that can help the company take action before a customer decides to go.

I used a real world dataset from IBM which has information on around 7000 customers. The dataset has details like how long the customer has been with the company, what kind of plan they are on, how much they pay every month, and whether they churned or not. This gave me a good base to do a proper analysis.

1.1 What is Churn?

Churn basically means when a customer stops using a service or product. So if someone was using Airtel for 2 years and then one day they switch to Jio and cancel their Airtel connection, that person churned. For the company, every churned customer means lost revenue. If the churn rate is too high it becomes a serious business problem.

Studies have shown that acquiring a new customer can cost 5 to 7 times more than retaining an existing one. That is why companies spend a lot of money trying to understand churn and prevent it. That's basically what this project is trying to do as well.

1.2 Project Scope

This project is focused on exploratory data analysis and building a few machine learning models to understand churn better. The work is structured into these main parts:

- Data loading and initial exploration
- Data cleaning and preprocessing
- Exploratory Data Analysis with visualizations
- Feature analysis to find what factors cause churn
- Applying machine learning algorithms for prediction

- Comparing model performances
- Final business recommendations based on findings

2. Dataset Overview

The dataset used in this project is the Telco Customer Churn dataset provided by IBM. It is freely available on Kaggle. The dataset has 7043 rows and 21 columns. Each row represents one customer and each column tells us something about that customer.

Below is a quick overview of the main columns in the dataset:

Column Name	Description	Type
customerID	Unique ID for each customer	Text
gender	Male or Female	Categorical
SeniorCitizen	Whether customer is a senior (1/0)	Numeric
tenure	How many months with the company	Numeric
Contract	Month-to-month, 1 year, 2 year	Categorical
MonthlyCharges	Amount charged per month	Numeric
TotalCharges	Total amount charged overall	Numeric
InternetService	DSL, Fiber optic, or None	Categorical
PaymentMethod	How the customer pays	Categorical
Churn	Yes = left, No = stayed	Target Variable

2.1 Class Imbalance

One thing I noticed right away when I loaded the data is that only about 26.5% of customers had churned and the remaining 73.5% had stayed. This kind of imbalance is quite common in churn datasets. It means we have to be careful when evaluating model performance because even if a model just predicts everyone stays, it would still get 73% accuracy which sounds good but is actually useless.

To handle this I made sure to look at other metrics like precision, recall and F1-score instead of just accuracy, which gives a more honest picture of how well the model is really doing.

3. Data Cleaning & Preprocessing

Before any analysis can be done, the data needs to be cleaned. Raw data is almost never perfect. I found a few issues in this dataset which needed to be fixed first.

3.1 Issues Found

- TotalCharges column — was stored as a text/object type instead of a number. This is because some rows had blank spaces instead of actual values.
- Missing values — after converting TotalCharges to numeric, 11 rows had NaN values, I filled them with 0.
- Churn column — It had Yes/No values, which needed to be converted to 1 and 0 for calculations.
- customerID — this column doesn't have any useful information for analysis so it was dropped.

3.2 Encoding Categorical Variables

For the machine learning part, all categorical columns like gender, Contract, PaymentMethod and InternetService had to be converted into numbers. I used two techniques for this:

- Label Encoding — for binary columns like gender (Male/Female) and Churn. (Yes/No)
- One Hot Encoding — for columns with more than 2 categories like Contract and InternetService, because you can't just assign numbers like 1,2,3 since that would imply some order which doesn't actually exist.

After encoding, the final dataset had around 30 features which were then used for model training.

3.3 Feature Scaling

Since the numeric columns like tenure, MonthlyCharges and TotalCharges have very different ranges, I applied StandardScaler to normalize them. This is important especially for algorithms like Logistic Regression and KNN which are sensitive to the scale of input features. Tree based models like Random Forest don't really need this but I scaled all features anyway to keep things consistent.

4. Exploratory Data Analysis (EDA)

This is probably the most important part of the whole project. EDA is basically about asking questions and letting the data answer them through charts and statistics. I made several visualizations to understand churn patterns across different customer segments.

4.1 Overall Churn Rate

The very first thing I checked was how many customers actually churned. Out of 7043 customers, 1869 had churned which is about 26.5%. This might not sound like a lot but for a telecom company with millions of customers, losing 1 in every 4 customers is a serious problem that can cost crores of rupees every year.

4.2 Effect of Tenure on Churn

Tenure means how long a customer has been with the company. When I plotted the distribution of tenure separately for churned and retained customers, the result was very clear. Customers who churned had an average tenure of just 18 months, while customers who stayed had an average tenure of around 38 months.

This tells us that the first few months are the most critical period. If a customer is not satisfied in the beginning, they will likely leave within 5-6 months. After that, if they are still with the company, the chances of them leaving become lower and lower over time.

4.3 Contract Type

This was probably the most interesting finding in the whole analysis. Month to month customers had a churn rate of around 43%, while customers on 1 year contracts had only 11% churn, and 2 year contract customers had just 3% churn. The difference is huge.

This makes sense intuitively also. If someone is on a monthly plan, they can cancel anytime without any penalty. But if they are on a yearly contract, they are committed. The data clearly shows that contract type is one of the strongest predictors of churn.

4.4 Monthly Charges

Customers who churned were paying on average around 74 dollars per month, while customers who stayed were paying around 61 dollars. So higher paying customers are

more likely to leave. This suggests that these customers might not feel they are getting enough value for the money they are spending, or they found a cheaper alternative.

4.5 Internet Service Type

Fiber optic internet users had a significantly higher churn rate (around 42%) compared to DSL users (around 19%) and customers with no internet service (only 7%). This was a bit surprising at first because fiber optic is usually the better and faster service. But it is also more expensive, which probably explains the higher churn among those customers.

4.6 Senior Citizens

Senior citizens had a churn rate of about 41% compared to 24% for non seniors. This is quite a big difference. One possible reason could be that seniors find it harder to deal with technical issues or complicated billing, and when they do have a problem and dont get proper support, they just switch to another provider.

5. Machine Learning Models

After the EDA part, I moved on to building machine learning models to actually predict which customers are likely to churn. The idea is that if we can predict churn before it happens, the company can take steps to retain those customers.

I tried four different algorithms and compared their results. The dataset was split into 80% training and 20% testing. I used stratified splitting to make sure both sets had similar proportions of churned and non churned customers.

5.1 Logistic Regression

I started with Logistic Regression as the baseline model. Its one of the simplest classification algorithms and works well when the relationship between features and the target is somewhat linear. The model gave a decent accuracy of around 80% but the recall for the churn class was not great, which means it was missing a fair number of churned customers.

Logistic Regression also gave us something useful, the feature coefficients which tell us which features are pushing the prediction toward churn and which ones are pulling away from it. Contract type, tenure and monthly charges came out as the most influential features, which matched what we found in the EDA.

5.2 Decision Tree

Next I tried a Decision Tree classifier. Decision trees are easy to understand and you can actually visualize what decisions the model is making at each step. However, without tuning they tend to overfit, meaning they perform very well on training data but not so well on test data.

I used max_depth parameter to limit the tree and prevent overfitting. With max depth of 5, the model gave around 78% accuracy. The advantage of decision trees is that they dont need feature scaling and handle categorical data well.

5.3 Random Forest

Random Forest is basically a combination of many decision trees. Each tree is trained on a random subset of the data and random subset of features, and the final prediction is made

by taking the majority vote across all trees. This helps reduce overfitting and usually gives better results than a single decision tree.

The Random Forest model gave the best overall performance with an accuracy of around 82% and a much better F1 score for the churn class. It also gave us feature importance scores which showed that tenure, total charges, monthly charges and contract type are the most important features, again consistent with the EDA findings.

5.4 K-Nearest Neighbors (KNN)

I also tried KNN with k=5. KNN works by finding the 5 most similar customers and predicting churn based on what those customers did. It gave about 78% accuracy. Its a simple and intuitive algorithm but can be slow on large datasets and doesnt handle class imbalance very well.

5.5 Model Comparison

Here is a comparison of all four models on the test set:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80.3%	66.2%	54.8%	59.9%
Decision Tree	78.1%	59.4%	52.1%	55.5%
Random Forest	82.1%	70.3%	62.4%	66.1%
KNN (k=5)	78.6%	61.7%	49.3%	54.8%

Based on these results, Random Forest gave the best overall performance. The F1 score is the most important metric here because it balances both precision and recall. A high precision means when we predict churn, we are usually right. A high recall means we are catching most of the actual churners.

6. Key Findings Summary

After doing the full analysis, these are the most important things I found:

- Month-to-month contract customers churn at 43%, compared to just 3% for 2-year contracts. Contract type is the **biggest churn factor**.
- New customers in the first 0-6 months have the highest churn risk. If a customer survives past 12 months, they are much less likely to leave.
- Higher monthly charges are strongly associated with churn. Customers paying above \$65/month have noticeably higher churn rates.
- Fiber optic internet users churn more than DSL users, despite having a better service. Cost seems to be the main reason.
- Senior citizens churn at almost double the rate of younger customers.
- Gender has almost no effect on churn. Male and female customers churn at nearly the same rate.
- Random Forest was the best performing model with 82.1% accuracy and 66.1% F1 score on the churn class.

7. Business Recommendations

Based on everything I found in this project, here are some practical things the telecom company can actually do to reduce churn:

#	Problem	Recommended Action
1	High churn on monthly plans	Give discounts or free upgrades to customers who switch to annual plans. Even a 10-15% discount can make a big difference.
2	New customers leaving early	Start an onboarding program. Call or email new customers in months 1, 2 and 3 to make sure they are happy and address any issues early.
3	High charges causing churn	Introduce loyalty discounts for customers paying above a certain amount. Give them a reason to stay.
4	Senior citizens churning more	Create a simplified plan for seniors with cheaper pricing and a dedicated helpline. Make the experience less complicated for them.
5	Fiber optic users leaving	Investigate if there are service quality issues. If not, consider adding more value to fiber plans (like free streaming subscriptions) to justify the cost.

These are not just guesses. Each recommendation is directly backed by what the data showed us. Targeting the right customer segment at the right time is the key to reducing churn effectively.

8. Conclusion

This project gave me a really solid understanding of how data science works in a real world business context. I started with a messy raw dataset and by the end I had clear, actionable insights that an actual company could use.

The EDA part was the most eye opening for me personally. I did not expect contract type to have such a massive effect on churn. The difference between monthly (43%) and 2-year (3%) contract churn rates is just huge, and without the analysis I wouldn't have known that.

On the machine learning side, I learned that accuracy alone is not a good metric when the classes are imbalanced. You have to look at precision and recall together, and the F1 score gives a nice single number that captures both. Random Forest was clearly the best model here, mainly because it is more robust and handles the complexity of the data better than simpler models.

If I had more time, I would have liked to try a few more things like XGBoost which generally performs even better than Random Forest on tabular data, and also SMOTE (Synthetic Minority Oversampling Technique) to handle the class imbalance more properly. But given the scope of this project, I think the results are quite solid.

Overall this project helped me connect the dots between data, analysis, models and real business decisions. That is what data science is really about at the end of the day.

References

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